

Windows Explorer allowing for quick and easy access.

Setting up back-ups

One of the most common and important uses for a NAS box is networked back-ups. With your newly set up NAS box, this is going to be extremely simple, provided you have Windows 7 Professional or later versions. To set things up, just open the 'Start' menu on Windows, and type 'backup' to get to the Windows 'Backup and Restore' utility. Click 'Set Up Backup' and then the 'Save on network' button. Proceed to enter the path to your NAS box manually, or browse to discover the server. Choose your back-up settings and set a schedule for your back-ups. I would recommend you connect your computer over Ethernet while the back-up is in progress, to speed things up. If you do not have Windows Professional, you can use an open source tool called Create Synchronicity for creating back-up images for your system.

For the Mac only

But what if all the computers in your house run the Mac OS, and you want to create a wireless Time Machine-based back-up solution for all your computers? Then it's time for a little more terminal magic. First of all, you need to install Avahi, which implements Apple's Bonjour protocol, and Netatalk, an open source implementation of the AppleTalk protocol that you will be using to share your drives with your Macintosh computers. Run the following commands in the terminal to download and install these services on your Linux system:

```
sudo apt-get install avahi-daemon
sudo apt-get install netatalk
```

Next, edit the configuration files to enable support for the latest version of Time Machine running on Lion or Mountain Lion. In the terminal, run:

```
sudo gedit /etc/netatalk/afpd.conf
```

Add the following line at the end of the file, and then save and close the file:

```
- tcp -noddp -uamlist uam_guest.so,uams_dhx2_
passwd.so -nosavepassword
```

Next, edit the *AppleVolumes* file to share your drives using AFP (Apple Filing Protocol)—run the following command in the terminal:

```
sudo gedit /etc/netatalk/AppleVolumes.default
```

Comment out the ~/ "Home Directory" entry at the end of the file by adding a 'hash' in front of it. Then add the following command below the commented line:

```
/path/to/mountedTimeMachine "UbuntuCapsule"
cnidscheme:dbd options:usedots,upriv,tm
allow:userName
```

Here, */path/to/mountedTimeMachine* is the address to the mount directories you created in Step 1, and *userName* is your Ubuntu username. Select the one you wish to be used as a Time Machine back-up. Next, add an entry for each disk below the Time Machine entry, to share your remaining drives over AFP as well. The entry will be in the following format:

```
/path/to/mountedMovieDisk "MediaCapsule"
cnidscheme:dbd options:usedots,upriv allow:userName
```

Make sure you have only one disk marked as the Time Machine back-up volume (specified by the 'tm' in the options). After you have added entries for all your disks, save this file and restart Netatalk with `sudo service netatalk restart`.

On your Mac, you can open up Finder, where your Ubuntu server should be visible in the left panel. Click on it, and authenticate with your username and password. All your shared disks should be visible under the server, including your Time Machine disk. Your Time Machine disk should also show up under Time Machine now, under the available disks. Select the disk and start backing up; it's as simple as that! You can select the same disk as a back-up destination for multiple Macs. Again, I would strongly recommend that you do the first full back-up over Ethernet. Time Machine does incremental back-ups subsequently, which should not be as large as the first one; those could be done over Wi-Fi, if desired.

Your media and back-up server is up and running now, and you can disconnect your display from it. Now that it is set up, there is a whole world of other things you can do with it. For starters, you could connect and share your printer over the network, so all your computers can access it—and there's one less item to clutter your workspace. You could also set up a Plex server to share and serve data to devices like your phone and tablet as well. Just start playing around with and exploring the wealth of free and open source software available for Ubuntu, and you will be amazed at how handy your home server can turn out to be! 

Links

- [1] Ubuntu Live CD image: <http://www.ubuntu.com/>
- [2] How to create an Ubuntu USB disk image: <https://help.ubuntu.com/community/Installation/FromUSBStick>

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